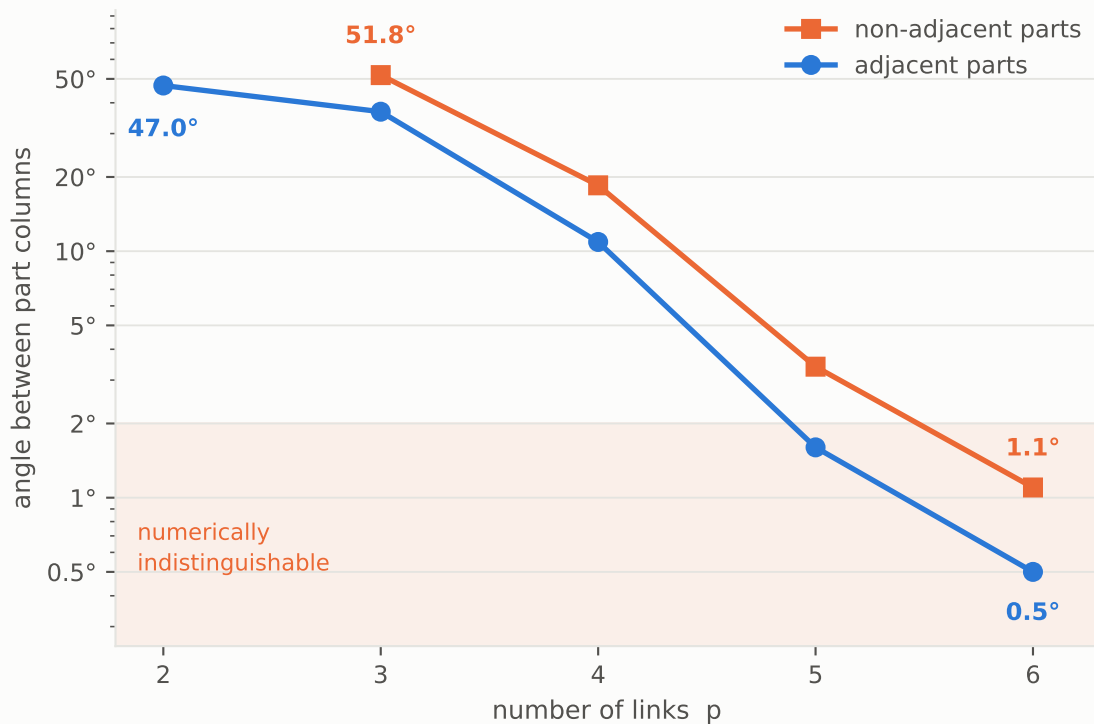


Part columns become parallel as the chain grows



Whitened regressor, mean over 40 random configurations. Parts differ only in the six torque rows that carry signal (Cor. 1), and that is exactly where angle error dominates the sensor noise — 22× at $p=3$, 41,690× at $p=4$. Whitening suppresses those rows, leaving the force rows, which were parallel to begin with.